EV Battery Course — Module 11: Battery Testing, Validation and Quality Engineering

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Battery quality is decided at micron scale and proven at pack scale: a 50 µm metal particle in a coating can become a field fire years later. This module maps every test from incoming material to end-of-line, classifies each as destructive or not, production or validation, and regulatory or internal, and then applies the quality tools you already know (MSA, SPC, Cp/Cpk, FMEA, root cause analysis) to battery manufacturing.

1. Learning objectives

- Lay out the test chain: incoming material → electrode → cell → module/pack → vehicle.
- Explain capacity, DCIR, impedance spectroscopy, self-discharge, cycle and calendar life, and abuse tests.
- Classify tests as destructive/non-destructive, production/validation, regulatory/internal.
- Apply MSA, SPC, Cp/Cpk, PFMEA and RCA to battery processes.
- Build a sample control plan for a battery line.
- Recognise real defect types and how each is detected.

2. Fundamentals: three kinds of testing

Kind	Purpose	Who / when	Sample
Development and validation (DV/PV)	Prove the design meets requirements and regulations	Engineering; before SOP and after changes	Samples, often destroyed
Production (in-process, EOL)	Prove each unit was built right	Manufacturing; every unit or sampling	100% or statistical
Field monitoring	Catch problems in use	Quality and service; through life	Fleet data, returns

Your PPAP/APQP experience maps directly: DV and PV testing feed the PPAP, the control plan governs production testing, and field data closes the loop.

3. Test catalogue

3.1 Incoming material inspection

Material	Key checks
Cathode / anode powders	Particle size distribution (laser diffraction), specific surface area (BET), moisture (Karl Fischer), tap density, impurities (ICP-OES: Fe, Cu, Zn, etc.), magnetic foreign matter, first-cycle capacity in coin cells
Electrolyte	Moisture (ppm), HF/acidity, composition (GC), colour
Separator	Thickness, porosity, air permeability (Gurley), shrinkage at temperature, puncture strength, coating adhesion
Foils	Thickness, tensile strength, surface cleanliness, pinholes
Cans, lids, tabs	Dimensions, burrs, weldability, coating

3.2 Electrode quality

- Coating weight (areal loading, mg/cm²) by beta-ray or X-ray gauges, online across width and length.
- Thickness and density after calendering (laser or contact gauges).
- Defect detection by line-scan cameras: pinholes, streaks, agglomerates, edge waviness, uncoated areas.
- Adhesion (peel test), resistivity, moisture after drying.
- Burr height after slitting and notching (microscope; often < 10 µm spec).

3.3 Cell tests

Test	What it reveals	Type
Hi-pot / insulation (dry cell before filling)	Shorts through separator from particles or burrs	Production, 100%
X-ray / CT	Electrode alignment, anode overhang, tab position	Production (sample or 100%)

Test	What it reveals	Type
Leak test (helium)	Weld and seal integrity	Production
Formation data	First-cycle capacity, gas, early shorts	Production, 100%
Self-discharge (OCV drop over ageing, "K-value")	Micro-shorts, contamination	Production, 100%
Capacity and energy	Grading	Production, 100%
ACIR (1 kHz) and DCIR (pulse)	Resistance, welding defects	Production, 100%
EIS (impedance spectroscopy)	Separates ohmic, charge-transfer and diffusion resistance	Development, failure analysis
Cycle life (temperatures, C-rates, DOD)	Fade rate	Validation, destructive over time
Calendar life (SOC × temperature matrix)	Storage fade	Validation
Abuse: overcharge, external short, crush, nail penetration, thermal (oven)	Safety behaviour	Validation / regulatory, destructive

3.4 Module and pack tests

Test	Type
Vibration (random, sine), mechanical shock	Validation / regulatory
Crush, mechanical integrity	Regulatory (e.g. UNECE R100, AIS-038)
Thermal shock and cycling, humidity	Validation
Water immersion, IP67/IP6K9K	Validation; leak test is production
Fire resistance (external fire)	Regulatory
Thermal propagation (single-cell runaway trigger)	Regulatory (GB 38031; Indian AIS amendments)
Overcharge, overdischarge, overcurrent, over-temperature protection	Validation / regulatory
External short circuit	Regulatory

Test	Type
EMC	Validation
Insulation resistance and dielectric withstand (hi-pot)	Production 100%
Leak test of enclosure and coolant loop	Production 100%
EOL functional (BMS comms, cell voltages, temperatures, contactors, pre-charge, HVIL, isolation monitor, current sensor calibration)	Production 100%

Destructive vs non-destructive. Abuse, crush, fire, propagation, nail, cycle-to-failure and teardown tests are destructive. Everything on the production line must be non-destructive (or use a separate sample).

Main standards: IEC 62660-1/2/3 (cells for EVs; Indian adoption IS 16893), UN 38.3 (transport), UNECE R100 (REESS safety), ISO 12405 (pack test procedures), ISO 6469-1, SAE J2464 / J2929, GB 38031 (China), AIS-038 Rev 2 (cars) and AIS-156 (2W/3W) in India. Always work from the current revision.

4. Diagram: test gates across the value chain

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flowchart LR
  A["Incoming material<br/>PSD, moisture, impurities"] --> B["Electrode<br/>coating weight, defects, burrs"] --> C["Cell assembly<br/>hi-pot, X-ray, leak"] --> D["Formation and ageing<br/>capacity, self-discharge, DCIR"] --> E["Grading<br/>bin by capacity, resistance"] --> F["Pack EOL<br/>insulation, leak, BMS, functional"] --> G["Vehicle EOL<br/>HV checks, charging test"]
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5. Step-by-step: how a hidden metal particle becomes a field failure, and where it is caught

1. A 50 µm steel particle from worn equipment enters the cathode slurry. → *Catch:* magnetic separators, ICP, foreign-matter checks.
2. It is coated into the electrode. → *Catch:* vision inspection (sometimes).
3. Calendering presses it toward the separator. → *Catch:* hi-pot test on dry cell.
4. In formation, iron dissolves at cathode potential and redeposits on the anode, growing a dendrite. → *Catch:* abnormal formation voltage.

5. During ageing, a micro-short slowly discharges the cell. → *Catch:* **self-discharge (K-value) screening**.
6. If missed: in the field, the micro-short grows with cycling and heat, causing local heating and possibly thermal runaway.

This is why self-discharge screening, despite days of work-in-progress, is one of the most important tests in cell manufacturing.

6. Engineering equations

Process capability:

$$C_p = \frac{USL - LSL}{6\sigma} \qquad C_{pk} = \min\left(\frac{USL - \mu}{3\sigma},\ \frac{\mu - LSL}{3\sigma}\right)$$

Measurement system (gauge R&R as % of tolerance):

$$\%GRR = \frac{6\,\sigma_{GRR}}{USL - LSL} \times 100$$

Self-discharge rate (K-value):

$$K = \frac{OCV_{1} - OCV_{2}}{t_2 - t_1} \quad (\text{mV/day})$$

DCIR from a pulse:

$$R_{DC} = \frac{\Delta V}{\Delta I}$$

Pressure-decay leak rate:

$$Q = \frac{V\,\Delta p}{\Delta t}$$

Risk priority number (legacy FMEA):

$$RPN = S \times O \times D$$

7. Worked examples

Example 1: Cp/Cpk for coating weight. Spec 20.0 ± 0.6 mg/cm²; process mean 20.15, σ = 0.12.

- Cp = 1.2 ÷ (6 × 0.12) = **1.67**
- Cpk = min[(20.6 − 20.15) ÷ 0.36, (20.15 − 19.4) ÷ 0.36] = min(1.25, 2.08) = **1.25**

- Capable process but off-centre; re-centring the mean to 20.0 brings Cpk to 1.67. Automotive customers typically require Cpk ≥ 1.33 for important characteristics and ≥ 1.67 for safety-critical ones.

Example 2: gauge R&R on a DCIR tester. Tolerance 0.20–0.30 mΩ (0.10 mΩ wide); σ_GRR = 0.0025 mΩ.

- %GRR = 6 × 0.0025 ÷ 0.10 = **15%**: marginal (10–30% conditionally acceptable under AIAG MSA). Improve fixtures, contact resistance and temperature control.

Example 3: self-discharge screening. OCV measured after 2 days and 9 days of ageing at 25 °C.

Cell	OCV day 2 (mV)	OCV day 9 (mV)	K (mV/day)	Decision
A	3,331.4	3,330.0	0.20	Pass
B	3,331.2	3,329.7	0.21	Pass
C	3,331.5	3,322.1	**1.34**	Reject: likely micro-short

Limits are set statistically from the population (e.g. mean + several σ) because LFP's flat curve makes absolute values tiny.

Example 4: pack leak test. Free air volume 100 L (0.1 m³); test pressure 5 kPa; observed drop 50 Pa in 60 s.

- Q = 0.1 × 50 ÷ 60 = 0.083 Pa·m³/s ≈ **0.83 mbar·L/s**. Compare with the limit correlated to IP67 water-ingress tests; temperature changes during the test must be compensated.

Example 5: PFMEA line (legacy RPN). Failure mode: laser weld undersized on busbar. Effect: high resistance, hot spot, possible fire (S = 9). Cause: lens contamination (O = 4). Current control: visual check (D = 6) → RPN = 216. Action: add post-weld resistance measurement and power monitor (D → 2) → RPN = 72. Under the AIAG–VDA FMEA method, a severity of 9 with these ratings would be High action priority regardless of RPN.

8. Sample control plan (abridged) for an LFP prismatic cell and pack line

Process step	Characteristic	Spec / target	Method	Frequency	Reaction plan
Incoming cathode	Moisture	≤ 500 ppm (example)	Karl Fischer	Each lot	Quarantine lot; supplier 8D
Slurry mixing	Viscosity, solid content	Recipe window	Viscometer, oven	Each batch	Hold batch; adjust

Process step	Characteristic	Spec / target	Method	Frequency	Reaction plan
Coating	Areal weight	20.0 ± 0.6 mg/cm²	Beta gauge, online	Continuous + SPC	Auto-mark; stop if out of control
Drying	Residual NMP/moisture	Below limit	Sample analysis	Each roll	Re-dry or scrap
Calendering	Thickness / density	±2 µm (example)	Laser gauge	Continuous	Adjust gap; segregate
Slitting / notching	Burr height	< 10 µm	Microscope	Start of roll, every N m	Change blade; hold
Winding / stacking	Overhang, alignment	⩾ 0.5 mm overhang	CCD vision / X-ray	100%	Reject
Tab welding	Pull strength, resistance	Min pull force	Pull test (sample), resistance (100%)	Per shift / 100%	Adjust; containment
Dry cell	Insulation	No breakdown at test voltage	Hi-pot	100%	Reject
Sealing	Leak rate	Below limit	Helium leak	100%	Reject
Filling	Electrolyte mass	± 1%	Weighing	100%	Reject/rework
Formation	Capacity, voltage curve	Within window	Cycler data	100%	Reject; analyse
Ageing	K-value	Statistical limit	OCV tester	100%	Reject
Grading	Capacity, DCIR	Bin limits	Tester	100%	Bin / reject
Pack: busbar welding	Joint resistance	⩽ limit	Micro-ohmmeter	100%	Rework / scrap
Pack: TIM	Bead weight / coverage	Window	Scale, vision	100%	Rework
Pack: sealing	Leak rate	Below limit	Pressure decay	100%	Rework seal
Pack EOL	Insulation, HVIL, BMS, contactors	Spec	EOL tester	100%	Repair loop

9. Typical defects and how they are detected

Defect	Cause	Risk	Detection
Coating non-uniformity	Slot-die wear, slurry variation	Capacity spread, plating at thin/thick spots	Beta gauge mapping, SPC

Defect	Cause	Risk	Detection
Burrs on edges	Worn blades, laser parameters	Separator puncture, short	Microscope, hi-pot
Metallic contamination	Equipment wear, environment	Micro-shorts, fires	Magnetic separation, self-discharge screening
Poor welding	Contamination, parameter drift	Resistance, hot spots, open circuit	Pull test, resistance, weld monitoring
Electrolyte leakage	Seal or weld defects	Corrosion, ageing, safety	Helium leak test, smell/visual
Capacity variation	Coating weight, formation variation	Imbalance, reduced pack capacity	Grading, matching
Moisture contamination	Dry-room excursion	Gas, HF, fast ageing	Dew-point monitoring, gas in formation

10. Root cause analysis and traceability

- **Traceability:** every cell has a laser-marked ID linking material lots, electrode rolls, machine parameters, formation data and grading results; packs link to cell IDs and BMS serial. A field fire can then be traced to a coating roll or a weld station shift.
- **RCA tools:** 8D, 5-Why, Ishikawa (fishbone), fault tree analysis, Shainin; supported by teardown, CT scanning, SEM/EDS and data analytics on production records.
- **Supplier quality:** PPAP, process audits (VDA 6.3), incoming quality data, change notification control. Cell suppliers changing a material is a major change requiring revalidation.

11. Real-world EV examples

- **Chevrolet Bolt recall (2020–2021):** LG cells from specific plants had two rare defects together (torn anode tab and folded separator), causing fires; GM recalled about 140,000 vehicles and replaced packs. It shows how rare cell-level defects can cost billions.
- **Hyundai Kona Electric recall (2020–2021):** linked to cells from LG with possible folded anode tab; about 82,000 vehicles globally had packs replaced.
- **Indian e-scooter fires (2022):** led to an expert investigation and the tightened AIS-156 amendments for cells, BMS and pack design.

12. Manufacturing applications (cost of quality)

- Formation and ageing tie up inventory for days to weeks; shorter ageing saves cost but raises escape risk: a key trade-off engineers must quantify.
- Each percent of yield lost in cell manufacturing is expensive because most of the value (materials) is already committed.
- Inline sensors and data analytics (machine learning on formation curves) are increasingly used to predict bad cells earlier.

13. Common mistakes and misconceptions

- **"Passing UN 38.3 means a cell is safe in an EV."** UN 38.3 is a transport test; vehicle safety needs much more.
- **"Capacity test is enough."** Many dangerous defects show only as self-discharge or impedance anomalies.
- **"Cpk from 30 parts proves capability."** Short-term studies need confirming with long-term data.
- **"The MSA is fine because the gauge is calibrated."** Calibration ≠ repeatability and reproducibility, especially with contact resistance at milliohm level.

14. Practical case study: rising self-discharge rejects

Problem. K-value rejects rise from 0.3% to 1.8% over two weeks.

Investigation. Rejects cluster on one winding machine and night shift. Teardowns show metallic particles; EDS identifies stainless steel. A guide roller bearing on the winder was wearing and shedding particles.

Containment. Quarantine and 100% extended ageing for cells from that machine; check all shipped cells by traceability.

Corrective action. Replace bearing; add particle-shedding inspection to preventive maintenance; install a particle counter near the winder; add SPC on K-value reject rate per machine.

Lesson. Traceability plus per-machine SPC turned a field-risk into an internal cost.

15. Revision notes

- Test classes: validation (prove design), production (prove each unit), field (prove in use).
- Self-discharge screening catches micro-shorts that capacity tests miss.

- Cpk ⩾ 1.33 typical, ⩾ 1.67 for safety-critical; %GRR < 10% good, 10–30% marginal.
- Every cell traceable to its materials and process data.
- Regulatory: AIS-038 Rev 2, AIS-156, IS 16893 / IEC 62660, UN 38.3, UNECE R100.

16. Glossary

Term	Meaning
K-value	Self-discharge voltage-drop rate
EIS	Electrochemical impedance spectroscopy
ACIR / DCIR	AC (1 kHz) and DC pulse resistance
Hi-pot	High-voltage insulation test
PFMEA	Process failure mode and effects analysis
GRR	Gauge repeatability and reproducibility
8D	Eight-discipline problem-solving method
SOP	Start of production

17. Knowledge check

Conceptual

1. Why can't production testing include nail penetration?
2. Why is self-discharge screening so slow, and why do factories keep it?
3. What does EIS reveal that DCIR does not?
4. Why is traceability essential for battery recalls?
5. Why is MSA particularly hard for milliohm-level measurements?

Numerical

1. Spec 3.300 ± 0.010 V; μ = 3.304 V, σ = 0.002 V. Compute Cp and Cpk.
2. A cell's OCV drops from 3,330.0 mV to 3,327.2 mV in 7 days. K-value?
3. A pulse of 100 A drops voltage by 28 mV. DCIR?

Application

1. Draft the five PFMEA lines you would prioritise for a pack assembly line.
2. A supplier proposes reducing ageing time from 14 to 7 days. What data would you require?

18. Answer key

Conceptual

1. It destroys the cell; production tests must be non-destructive.
2. Micro-shorts cause tiny voltage drops that need days to measure above noise. Catching them prevents field failures, which cost far more.
3. It separates resistance into ohmic, charge-transfer and diffusion components by frequency, helping diagnose which mechanism changed.
4. It lets you limit a recall to affected lots and vehicles and find the root cause.
5. Contact resistance, thermal EMFs and temperature effects are similar in size to the measured value; fixtures and technique dominate variation.

Numerical

1. Cp = 0.020 ÷ 0.012 = **1.67**; Cpk = min[(3.310 − 3.304)/0.006, (3.304 − 3.290)/0.006] = min(1.0, 2.33) = **1.0** (not capable; re-centre).
2. 2.8 ÷ 7 = **0.40 mV/day**.
3. 0.028 ÷ 100 = **0.28 mΩ**.

Application

1. Busbar weld undersized (hot spot); torque on HV connections out of spec; TIM void (hot cell); enclosure seal defect (water ingress, isolation fault); BMS harness misrouted or pinched (wrong cell reading or short). Rate S/O/D, define controls (resistance, torque monitoring, bead weight/vision, leak test, EOL cell-voltage plausibility).
2. Distribution of K-values at 7 vs 14 days on large samples; detection rate of known defective cells at each duration; correlation with field returns; statistical limits and false-accept risk; extra controls (e.g. elevated-temperature ageing) that compensate.

19. Further learning

- AIAG–VDA *FMEA Handbook* (2019); AIAG *MSA* (4th ed.) and *SPC* manuals.
- IEC 62660-1/2/3; UN Manual of Tests and Criteria, section 38.3; UNECE R100.
- AIS-038 Rev 2 and AIS-156 with amendments (ARAI website).
- Kwade et al., "Current status and challenges for automotive battery production technologies," *Nature Energy* 2018.

Next: Module 12, the lithium-ion manufacturing process.